

AI 101 Glossary

Abundant Free-Flowing Intelligence: A potential future scenario where AI computing power is limited only by energy and computational power, leading to unprecedented prosperity and progress

Agentic AI: AI systems characterized by their ability to act autonomously and collaborate with other agents to achieve goals

AI Agents: Intelligent software entities that can perform tasks autonomously and interact with their environment

AI Models: The underlying algorithms and architectures that power AI applications, trained on vast amounts of data to perform specific tasks

AI-Powered Robotics: The integration of intelligent machines into daily life, working alongside humans in various settings

Algorithm: A set of step-by-step instructions or rules a computer follows to complete a task or solve a problem

Application Programming Interface (API): A set of protocols and tools for building software applications, allowing different systems to communicate and exchange data

Artificial General Intelligence (AGI): A hypothetical future AI system with human-like or superhuman intelligence across a wide range of domains

Artificial Intelligence (AI): The field of computer science focused on creating systems that can perform tasks that normally require human intelligence

Bayesian Statistics: An approach to data analysis and parameter estimation based on Bayes' theorem

Bias in AI: Systematic errors in AI systems that can arise from flaws in the data or the algorithms themselves

CIDIA Circle: A framework describing five AI superpowers – create, interpret, discover, interface, and automate

Chatbot: A computer program designed to simulate conversation with human users

Computer Vision: The ability of computers to interpret and understand visual information from the world, such as images or videos

Context Size: The amount of additional information (e.g., documents, PDFs, links) that can be fed into an AI system to provide context for better understanding and performance

Conversational Interfaces: User interfaces that allow humans to interact with AI using natural language, as exemplified by ChatGPT

Custom GPTs: GPT models fine-tuned on specific domains or tasks to provide tailored outputs

Deep Learning: A type of machine learning that uses neural networks with many layers to analyze complex data, like images or speech

Digital Twin: A virtual representation of a person, object, or system that can be used for various purposes, such as simulation, monitoring, or optimization

Digital Utopians: People who believe that digital technologies have the potential to revolutionize society and play down technological risks

EU AI Act: The European Union's comprehensive legal framework for AI, categorizing AI systems based on their risk levels and setting guidelines for their development and deployment

Edge Computing: Running AI models on local devices like computers or phones, rather than in the cloud

Generative AI: AI models capable of creating new content, such as text, images, or music

GitHub Copilot: An AI-powered code completion tool that helps developers write code faster

Generative Pre-Trained Transformer (GPT): an advanced AI model developed by OpenAI for natural language processing, capable of understanding and generating human-like text based on a large corpus of pre-trained data using the Transformer neural network architecture

GPT-4: An advanced large language model developed by OpenAI, known for its strong performance across a wide range of tasks

Human-Assisted Design: A paradigm shift where AI takes a proactive role in the creative process, with humans acting as curators, guides, and visionaries

Large Language Model (LLM): An AI model trained on massive amounts of text data, designed to understand and generate human-like language

Machine Learning (ML): A subset of AI where computers learn from data and improve their performance on a task without being explicitly programmed

Model: A mathematical representation of what a machine learning algorithm has learned from training data

Model-Incorporated Data: AI models that have built-in knowledge from their training data

Multi-Agent Systems: Collaboration between multiple AI agents with different capabilities to solve complex problems more efficiently than humans

Multimodal AI: AI models that can understand and generate content across multiple modalities, such as text, images, video, and speech

Multimodality: The blurring of boundaries between speech, text, and imagery in AI systems, allowing for seamless conversion between different formats

Natural Language Processing (NLP): The ability of computers to understand, interpret, and generate human language

Neural Network: A computing system inspired by the biological structure of the brain, used in many AI applications to recognize patterns and make predictions

Open-Source Models: AI models that are freely available for anyone to use, modify, and distribute

Orchestrators: Tools that simplify the process of switching between different AI models based on specific requirements

Overfitting: A problem where a model learns the training data too well, leading to poor performance on new data

Prompt Engineering: The craft of communicating with AI models through carefully designed prompts to elicit desired outputs and behavior

Proprietary Models: AI models developed and owned by specific companies, often kept private and not shared with the public

Reinforcement Learning: A type of machine learning where an AI agent learns by interacting with an environment and receiving rewards or penalties

Retrieval-Centric Generation: AI models that can be fed with various types of documents (e.g., PDFs, videos, links) to provide context for better understanding and performance

Techno Skeptics: People who express doubt or caution regarding the potential benefits and risks of artificial intelligence

Text-to-Image Models

Text-to-Text Models: AI models that can understand and generate human-like text, handling a wide range of tasks. Examples include GPT-4 and Claude

Text-to-Video Models: AI models that can create videos based on textual input, such as OpenAI's Sora

Training Data: The information used to teach a machine learning model how to perform a task

Turing Test: A famous test of a machine's ability to exhibit intelligent behavior equivalent to, or indistinguishable from, that of a human